Shri Vile Parle Kelavani Mandal's
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC Accredited with "A" Grade (CGPA : 3.18)

Department of Computer Science and Engineering(Data Science)

SMASHFIX : AN APPLICATION FOR BADMINTON POSTURE CORRECTION USING DEEP LEARNING

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Introduction

- Computer vision and AI have revolutionized sports analysis by enabling precise, real-time tracking of player performance and biomechanics.

- In badminton, where footwork, stroke technique, and positioning are highly complex, these systems provide accuracy beyond human observation.

- Automated computer-vision-based systems can analyze posture and movement frame by frame, offering insights critical for player development.

- Deep learning integration allows precise detection of intricate body mechanics for training and competition analysis.

- Pose estimation algorithms map body positions using part affinity fields to study player movements in detail.

- Trajectory-based analyses track movement patterns to profile tactics and gameplay strategies.
- These methods surpass traditional observation by providing data-driven feedback for both technical and strategic improvement.

- Ongoing advancements in validation and system development continue to enhance badminton training and competitive performance.

Literature Survey

No.	Name of Paper	Conference / Journal (Year)	Methodologies	Parameters	Gaps
1.	Tennis player action dataset for human pose estimation	"Science Direct" (2024)	• Curation of Dataset using labelled data • COCO-Annotation used to highlight skeletal keypoints. • Vid frame extraction using smartphone's cam.	• 17 keypoints annotated using COCO-Annotator • Motion analaysis, Injury prevention an perf. Improvement.	• Lighting variations. • Camera and background noise. • Non -inclusion of more diverse players. • Lack of 3D pose estimation. • Lack of multiview vid data.
2.	Research on Badminton Teaching Technology Based on Human Pose Estimation Algorithm	"Machine Learning and Scientific Programing in Multi-Sensor Data Processing" (2022)	• Data Collection and Annotation. • Real-Time Feedback: • Convolutional Neural Networks (CNNs). • Multi-Person Pose Estimation. • Part Confidence Map.	• A comparison between two groups: one using the proposed method and the other using traditional teaching methods . • Contact point, stroke trajectory, stroke score. • Statistical analysis methods.	• Availability of dedicated datasets for badminton stance analysis • Constructed dataset only includes two skill movements.
3.	Application of Human Posture Recognition Based on the Convolutional Neural Network in Physical Training Guidance	"Computational Intelligence and Neuroscience" (2022)	• Convolutional Network Overview • Feature Map Size Maintenance • Stacked Hourglass Architecture • Large Receptive Field Module	• Addressing Errors in Complex Sports Video Environments • Superimposed Hourglass Grid. • The Need for Multimodal Approaches in Athlete Posture Estimation Technology	• Only estimates the pose of a single athlete in the sports game video and cannot solve the pose estimation problem of multiple athletes. • It has shortcomings in practical application and needs further research.

Literature Survey

No.	Name of Paper	Conference / Journal (Year)	Methodologies	Parameters	Gaps
4.	Machine Learning in Mobile Applications	"International Journal of Computer Science and Mobile Computing " (2022)	• **Server-Side ML:** Heavy computation is performed in the cloud. • **Client-Side ML:** Models run on the device for real-time inference, reducing latency. • ML frameworks such as TensorFlow, ML Kit, Pytorch used.	• Computation efficeincy (CPU/GPU Usage) • Srorage requirements • Network Latency, server side and client side repsonse time. • Scalability ,handling multiple requests and updating models	• Latency Issues in Server-Side ML, high infrastructure cost. • **Client-Side ML** – Model updates require app updates, making maintenance difficult. • Running ML models on mobile can **drain battery quickly.**
5.	Deep Learning Based Technical Classification of Badminton Pose with Convolutional Neural Networks	"ILKOM Jurnal Ilmiah" (2024)	• Two-Stage Pipeline in the Blaze-Pose Architecture. • Keypoint Regression Method.	• Confusion Matrix for Checking Model Performance. • Using f1-score, precision and recall.	• Limited dataset size and diversity for covering all badminton techniques and scenarios. • Model performance affected by outliers or unusual poses; requires further data preprocessing.
6.	Vision Based Automated Badminton Action Recognition Using the New Local Convolutional Neural Network Extractor	"Enhancing Health and Sports Performance by Design" (2020)	• Halved to reduce empty space and improve accuracy. • AlexNet used for feature extraction via transfer learning ("local CNN").	• Validation: Two confusion matrices used post Local CNN, AlexNet, and SVC. • Accuracy metric assesses improvement with Local CNN.	• Limited Actions; Only hit vs. non-hit. • Small Dataset; 2,990 images from 5 videos; no diversity.

Literature Survey

No.	Name of Paper	Conference / Journal (Year)	Methodologies	Parameters	Gaps
7.	Real-time Posture Correction of Squat Exercise: A Deep Learning Approach for Performance Analysis and Error Correction	"IEEE Access" (2025)	• **Stereo camera** captures squat videos. •**MediaPipe** extracts and tracks keypoints. •**Bi-CGRU** classifies squats. •**Deep learning regressor** estimates correct joint angles. •**Real-time feedback** compares incorrect and correct posture.	• Squat classification types • Joint angles measured • Dataset • Accuracy Metrics - Bi-CGRU model - Pose correction regression model	• Single-camera limitation • Pose estimation noise • Limited dataset diversity • No real-time mobile implementation • Lack of personalized feedback
8.	Enhancing Cricket Performance Analysis with Human Pose Estimation and Machine Learning	"Sensors" (2023)	•**Pre-processing** removes noise from videos. •**MediaPipe** extracts 17 keypoints for skeletal pose representation. •**Models tested:** RF, SVM, KNN, Decision Tree, LR, LSTM. •**RF performed best, LSTM performed worst.**	•Precision, Recall, F1 score • Acc : 97% , K fold : 95% with SD 0.07 ••Trained on various 3 different splits to entire generalization and get better understanding of performance • 80:20 gave best performance with RF	•**Lacked** defensive and modern reverse shots. •**Unbalanced dataset**, with more On Drive than Flick shots. •**PCA caused** minor data loss.
9.	Yoga Pose Detection and Correction using Posnet and KNN	"IRJET" (2022).	•**Images optimized** to uniform size. •**KNN classifier** detected poses via limb keypoints. •**Webcam** provided real-time feed. •**PoseNet** identified human poses using key points.	• Training Accuracy : 93.56% • Testing Accuracy: 98.51% • Testing using real time feed	• Very less poses used • OpenPose can be inaccurate during overlaps

Research Gaps

1. Most existing research focuses on pose estimation or movement classification, with limited emphasis on real-time corrective feedback.

2. Current correction-based systems (e.g., yoga applications) have limited scope in terms of the number of poses and are unsuitable for complex, fast-paced sports like badminton.

3. There is a lack of complete teaching aids that systematically teach or improve novice players’ badminton skills.

4. Limited mobile accessibility in existing sports training systems restricts user convenience and flexibility.

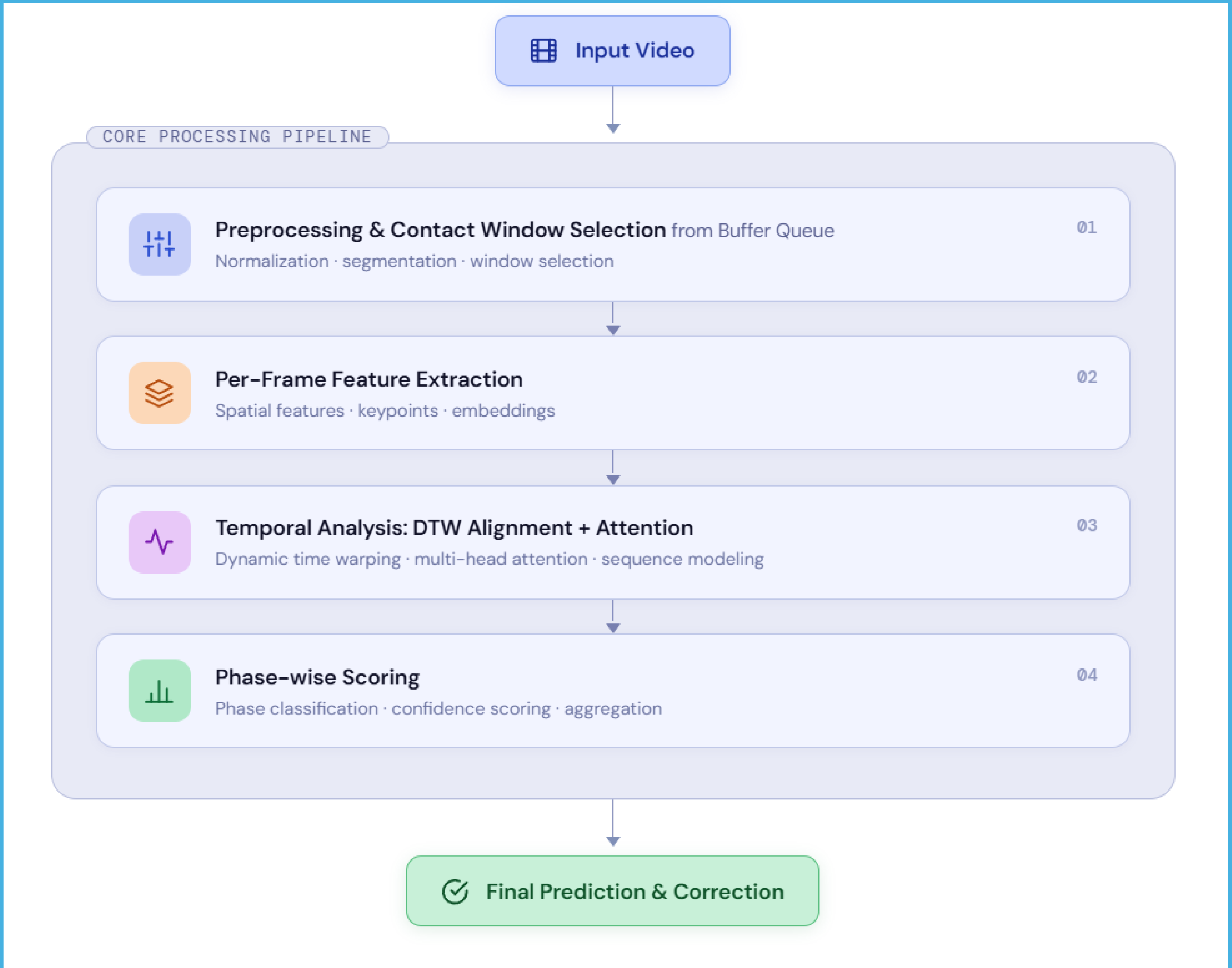
5. Most research models are limited to a small number of shot types and movement patterns in badminton analysis.

6. Many posture detection systems rely solely on client-side or server-side processing, leading to performance or latency issues.

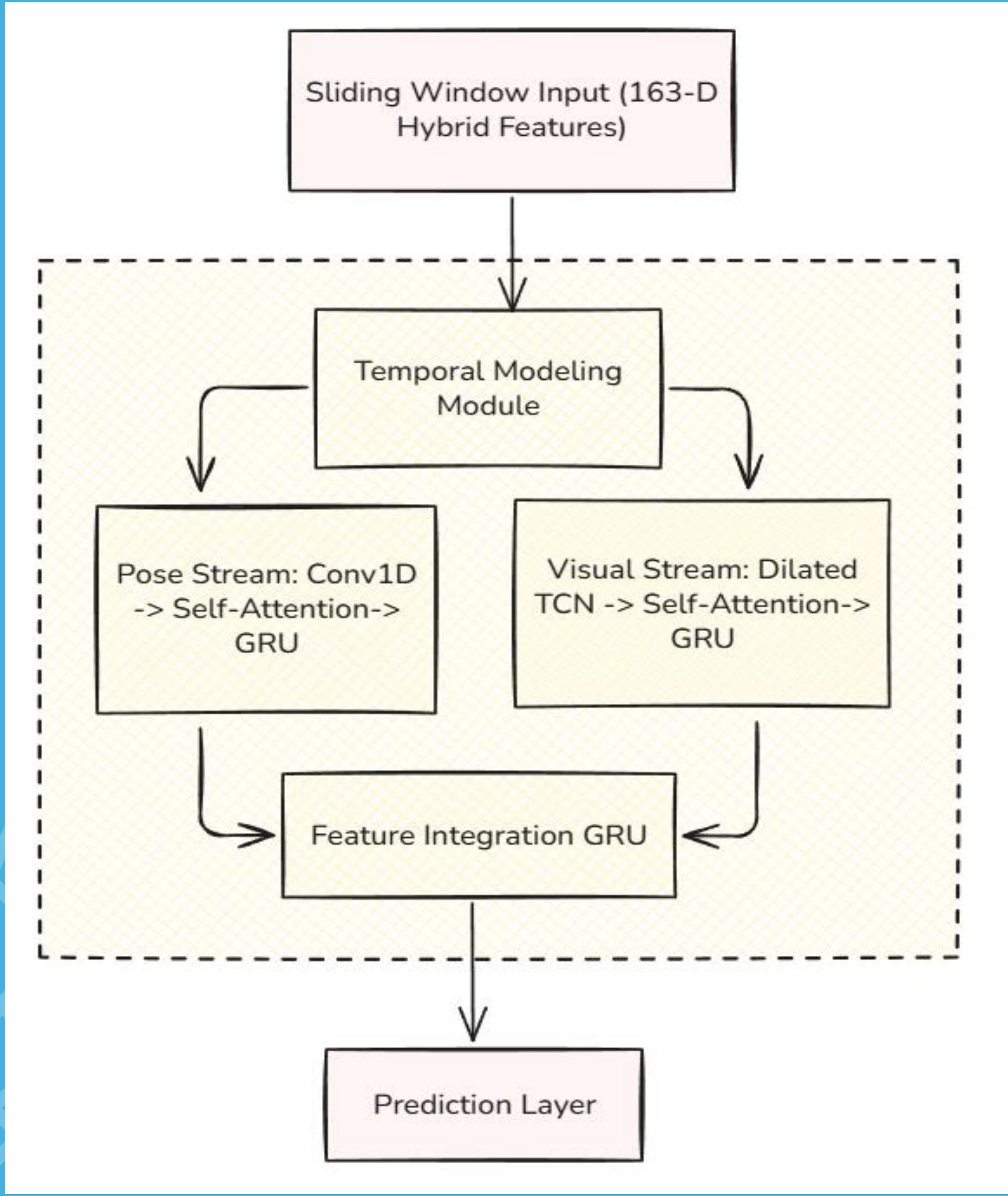
Problem Statement

Traditional badminton training relies heavily on manual coaching, which often misses subtle posture errors and limits real-time feedback, ultimately leading to inefficient skill development and a higher risk of injury. To overcome these issues, an automated computer vision-based system utilizes AI and pose estimation to provide precise, objective, and instant feedback. By detecting incorrect movements during practice and offering real-time visual or text cues, the system corrects techniques on the spot while tracking performance over time to build personalized training plans. Ultimately, bridging the gap between subjective coaching and data-driven precision ensures safer, more effective, and highly scalable player development.

System Architecture & Workflow



Model Architecture



Hybrid Dual-Stream Architecture The model simultaneously processes visual and biomechanical data to maximize shot recognition accuracy:

•**Visual Stream:** MobileNetV2 extracts 64D features from a cropped ROI, passing through a TCN and GRU to capture dynamics like racket and clothing motion.

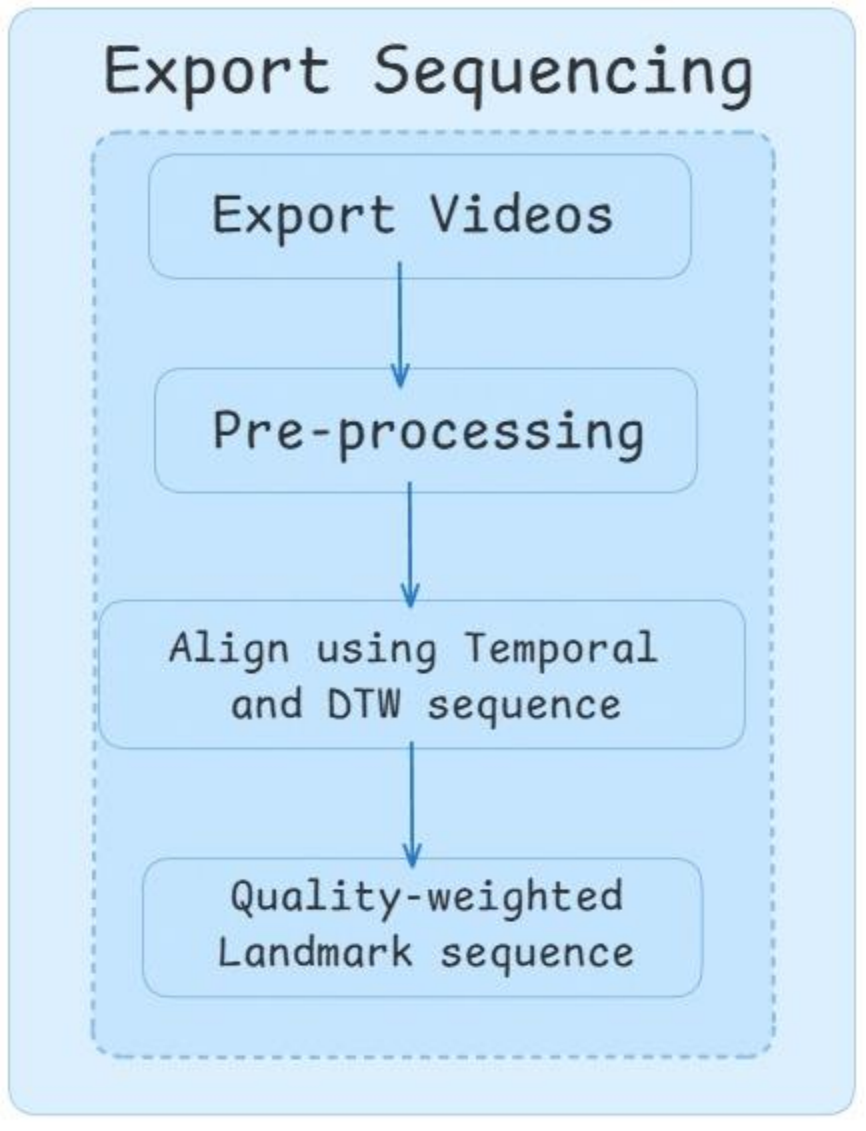
•**Pose Stream:** A dedicated GRU models 99D skeletal landmarks (33 joints) to track precise body mechanics, independent of background noise.

•**Fusion & Prediction:** Outputs from both streams merge into a final GRU, synthesizing context and mechanics to predict one of 6 shot classes.

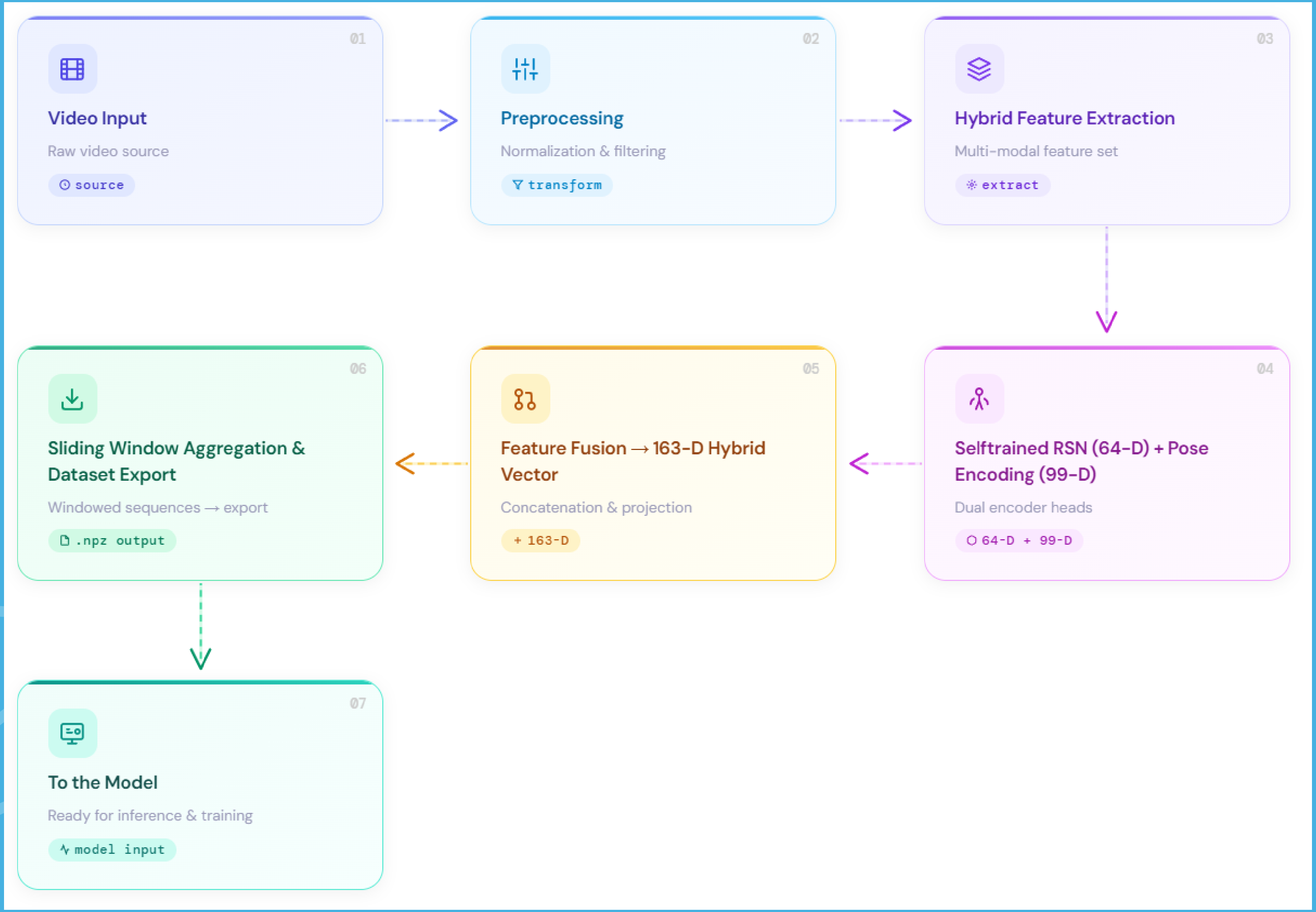
•**Sliding Window:** The system captures the full stroke progression by analyzing 40-frame sequences with a 5-frame stride.

Testing and Validation

- **Data Export & Pre-processing:** Expert videos are processed to extract raw landmarks and features.
- **DTW Alignment:** Uses **Dynamic Time Warping** to align multiple expert clips in time, ensuring disparate speeds of execution are synchronized.
- **Quality-Weighted Sequence:** A master template is generated by averaging aligned expert performances, creating a robust reference for scoring user errors.

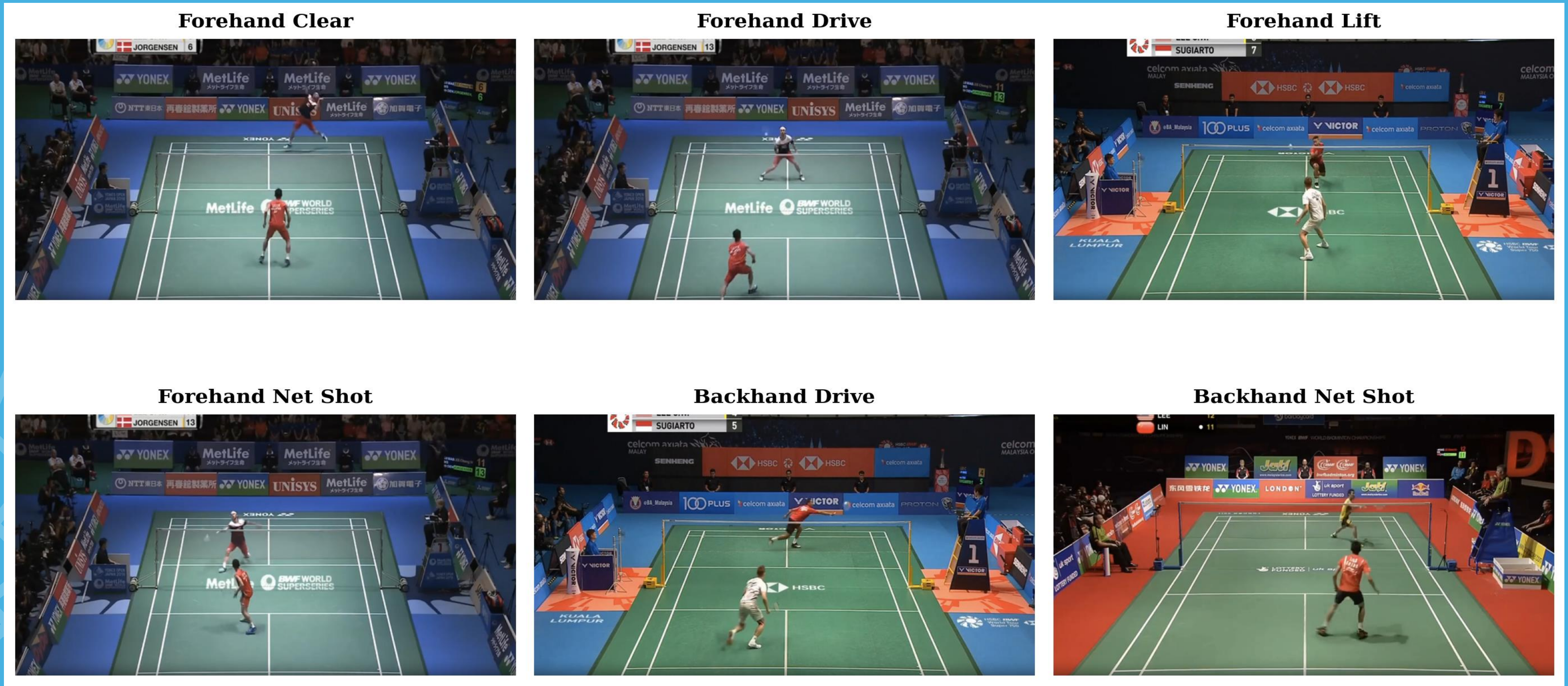


Results and output



Relevant Dataset

Access: Videos from BWF Indonesia Open
The "badminton_storke_video" dataset offers a collection of badminton stroke videos categorized into six distinct actions: **forehand drive, forehand lift, forehand net shot, forehand clear, backhand net shot, and backhand clear.**

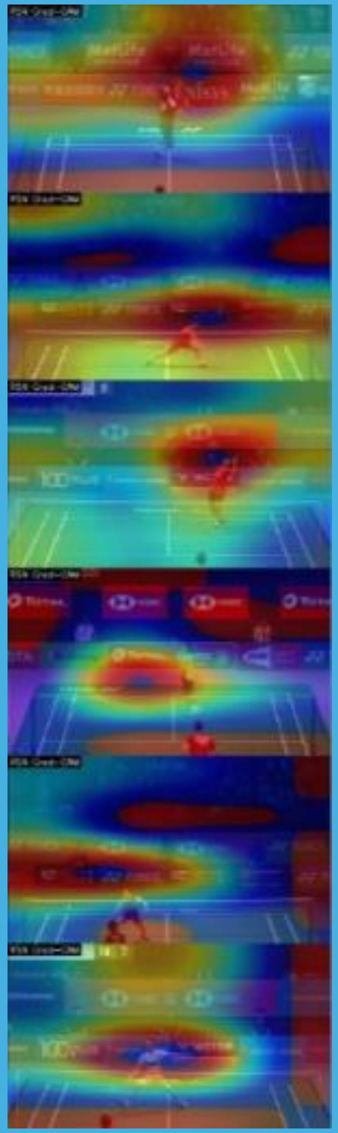


Demo Video

Correction Demo(Comparison)

Keypoints Extraction and Detection - 1

Keypoints Extraction and Detection - 2



Result Analysis

Conclusion & Future Scope

This project successfully developed Smashifix, an automated badminton coaching system that bridges the gap between amateur practice and professional biomechanical analysis. By utilizing a dual-pipeline architecture, the system delivers both ultra-low latency real-time feedback on the court and high-fidelity forensic analysis, achieving up to 98.92% validation accuracy. The integration of a mathematically rigorous scoring system with a Natural Language Coach ensures that complex movement data is seamlessly translated into actionable, personalized training drills. Ultimately, Smashifix provides an optimal balance of classification accuracy and feedback granularity, setting a strong foundation for future advancements like edge-device deployment and multi-player tracking.

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Dataset : Badminton Strokes Dataset